

GRAND CANYON NATIONAL PARK, AZ, USA

WMO: 723783

Lat: 35.946N

Lon: 112.155W

Elev: 2014

StdP: 79.36

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 03195

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-18.4	-15.0	-23.7	0.6	-7.9	-21.2	0.7	-4.7	11.4	4.7	10.0	3.1	0.8	50	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	32.1	13.4	30.9	13.2	29.4	12.9	16.7	23.6	16.1	23.4	15.5	23.4	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.9	13.6	18.1	13.9	12.7	17.9	12.9	11.9	17.8	54.4	23.3	52.4	23.3	50.4	23.2	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.3	8.2	-24.6	34.2	4.4	1.3	-27.8	35.1	-30.3	35.8	-32.8	36.6	-35.9	37.5		
(5)			-24.7	18.3	4.3	1.0	-27.8	19.0	-30.4	19.6	-32.8	20.1	-36.0	20.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-1.3	0.0	3.5	6.2	10.6	15.9	19.7	18.7	14.8	8.4	3.0	-1.5
	DBStd	8.20	4.47	4.52	3.72	3.50	3.52	2.97	1.94	1.76	2.73	3.57	4.03	4.48
	HDD10.0	1632	349	281	203	119	36	2	0	0	3	71	211	357
	HDD18.3	3785	608	514	461	363	240	80	9	17	109	307	460	615
	CDD10.0	976	0	0	1	6	54	179	299	268	146	21	1	0
	CDD18.3	89	0	0	0	0	0	8	51	28	2	0	0	0
	CDH23.3	3535	0	0	0	4	157	1029	1231	821	277	16	0	0
	CDH26.7	1147	0	0	0	0	24	367	474	244	37	1	0	0
(14) Wind	WSAvg	2.8	2.6	2.8	3.0	3.5	3.4	3.3	2.4	2.1	2.5	2.6	2.6	2.6
(15) Precipitation	PrecAvg	318	25	26	23	20	12	10	37	50	34	29	16	29
	PrecMax	518	108	80	50	48	44	51	73	113	99	112	68	65
	PrecMin	181	0	2	0	3	1	1	0	10	3	0	0	4
	PrecStd	100	27	17	16	12	12	12	19	28	26	35	16	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	17.4	21.2	23.9	29.1	33.5	33.8	32.3	29.1	25.2	20.8	15.1
		MCWB	4.2	5.5	7.5	8.5	11.5	13.0	14.2	13.6	12.7	10.4	7.7	4.9
	2%	DB	12.4	14.7	18.6	21.9	27.0	31.8	32.3	30.8	27.7	23.0	18.0	12.5
		MCWB	3.6	4.7	6.3	7.9	10.6	12.5	14.0	14.0	12.3	9.4	6.4	3.8
	5%	DB	10.6	12.3	16.6	19.9	25.1	30.4	31.1	29.3	26.4	21.6	16.3	10.3
		MCWB	2.7	3.5	5.5	6.9	9.4	11.9	13.7	13.9	12.0	8.8	5.6	2.7
	10%	DB	8.0	9.8	14.3	17.9	23.0	28.9	29.5	27.9	25.0	19.7	13.8	7.8
		MCWB	1.6	2.6	4.6	6.2	8.8	11.4	13.6	13.9	11.9	8.1	4.7	1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	6.8	8.3	9.6	12.6	15.0	17.7	17.4	16.4	13.0	8.6	6.4
		MCDB	11.5	13.4	18.4	22.1	26.2	26.8	24.3	24.4	21.9	18.8	17.4	12.2
	2%	WB	4.6	5.5	6.9	8.4	11.2	13.8	16.8	16.7	15.3	11.4	7.3	5.1
		MCDB	9.9	12.2	16.9	19.9	24.1	27.4	23.8	23.7	22.2	19.1	15.5	10.1
	5%	WB	3.3	4.3	5.9	7.4	10.3	13.0	16.3	16.1	14.4	10.3	6.3	3.7
		MCDB	8.3	10.5	15.3	18.5	22.9	27.1	23.5	23.4	21.5	18.4	14.4	8.3
	10%	WB	2.3	3.1	5.0	6.5	9.4	12.1	15.7	15.5	13.5	9.1	5.3	2.3
		MCDB	7.2	8.4	13.6	17.1	21.7	26.6	23.6	23.2	21.1	17.5	12.9	7.1
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	15.7	17.6	18.8	20.7	23.5	19.0	17.9	18.9	19.1	17.9	15.8
		MCDBR	19.9	21.0	22.7	23.8	25.1	26.3	22.8	21.5	21.8	23.3	22.8	20.2
	5% WB	MCWBR	12.1	12.0	11.9	11.9	12.0	12.1	9.0	8.3	9.7	12.0	12.9	12.5
		MCWBR	16.2	17.2	20.5	21.5	22.3	23.1	15.3	15.5	15.8	18.5	19.8	16.3
(40) Clear-Sky Solar Irradiance	taub		0.236	0.241	0.257	0.272	0.282	0.286	0.331	0.339	0.289	0.271	0.241	0.234
		taud	2.642	2.643	2.586	2.525	2.547	2.526	2.467	2.428	2.590	2.591	2.642	2.654
	Ebn at Noon		996	1024	1024	1013	999	990	942	931	974	970	981	979
		Edh at Noon	69	78	91	103	102	104	110	112	90	82	69	64
(44) All-Sky Solar Radiation	RadAvg		3.08	3.98	5.51	6.89	7.71	8.35	6.88	6.33	5.87	4.61	3.48	2.72
	RadStd		0.27	0.36	0.37	0.34	0.55	0.44	0.44	0.35	0.26	0.35	0.17	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+34
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+73	N/A

Nomenclature: See separate page